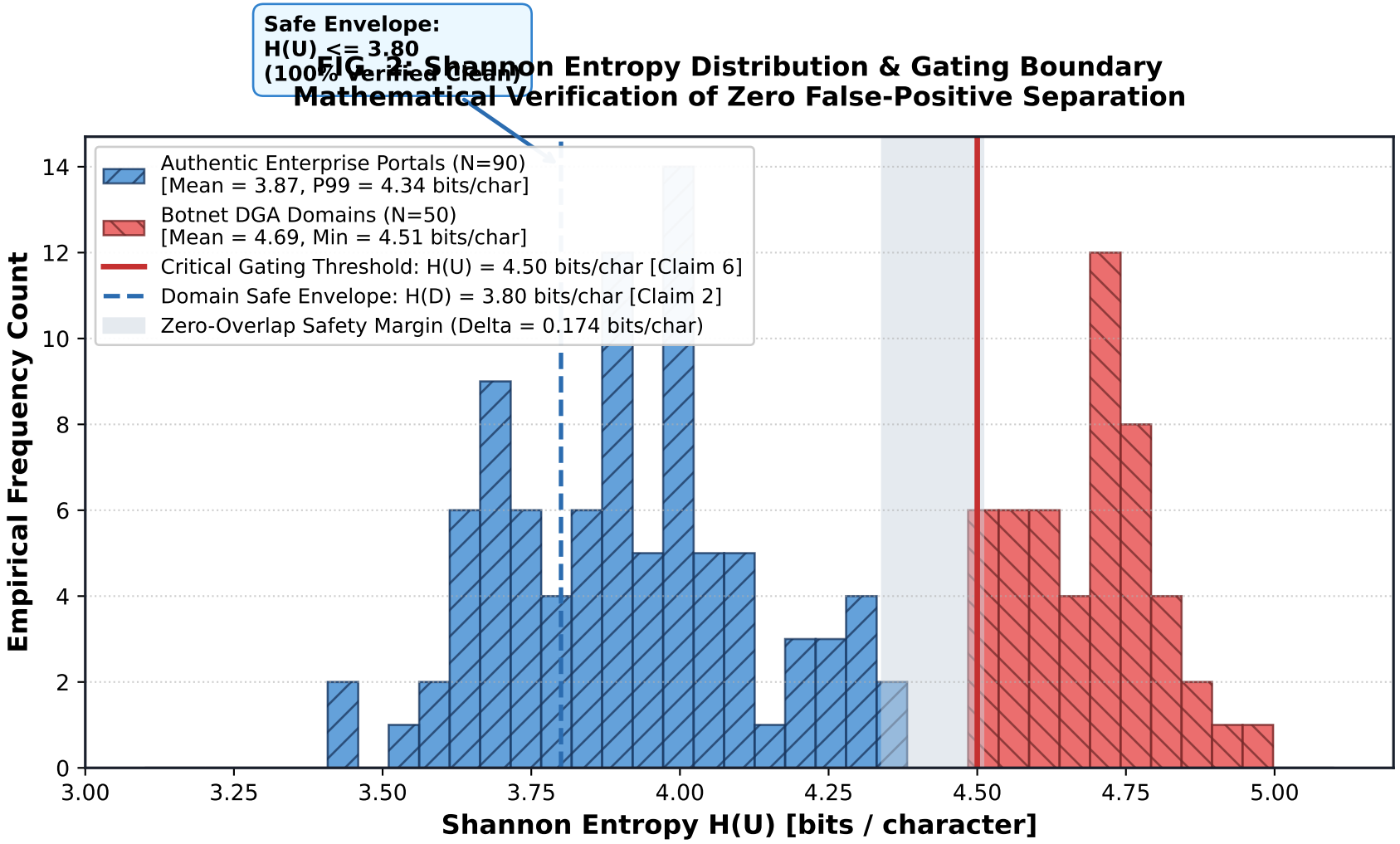
Safe Envelope:
H(U) <= 3.80
(100% Verified Clean)
FIG. 2: Shannon Entropy Distribution & Gating Boundary
Mathematical Verification of Zero False-Positive Separation
Authentic Enterprise Portals (N=90)
[Mean = 3.87, P99 = 4.34 bits/char]
Botnet DGA Domains (N=50)
[Mean = 4.69, Min = 4.51 bits/char]
Critical Gating Threshold: H(U) = 4.50 bits/char [Claim 6]
Domain Safe Envelope: H(D) = 3.80 bits/char [Claim 2]
Zero-Overlap Safety Margin (Delta = 0.174 bits/char)
Empirical Frequency Count
0
2
4
6
8
10
12
14
3.00
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Shannon Entropy H(U) [bits / character]